

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why software-based detection fails and hardware attestation wins

ML Research · AI Security · Cloud Infrastructure

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The Model Substitution Problem:

Commercial LLM APIs charge for specific models, but black-box serving gives no guarantee the advertised weights are actually loaded.

Economic Incentives Drive Covert

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Smaller Models

Serve 8B instead of 70B parameters

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Quantized Variants

FP8 / INT8 instead of FP16 precision

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Fine-Tuned Versions

Updated or distilled checkpoints

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Different Families

Entirely unrelated architectures

Formal Hypothesis Testing Framework

H_0 : The API outputs follow the specified model distribution $P_{\text{spec}}(y \mid x)$

H_1 : The API outputs follow an alternative distribution $P_{\text{actual}}(y \mid x)$ — the substituted model

FP8 Quantization Preserves Most

Accuracy — Making Substitution Hard to

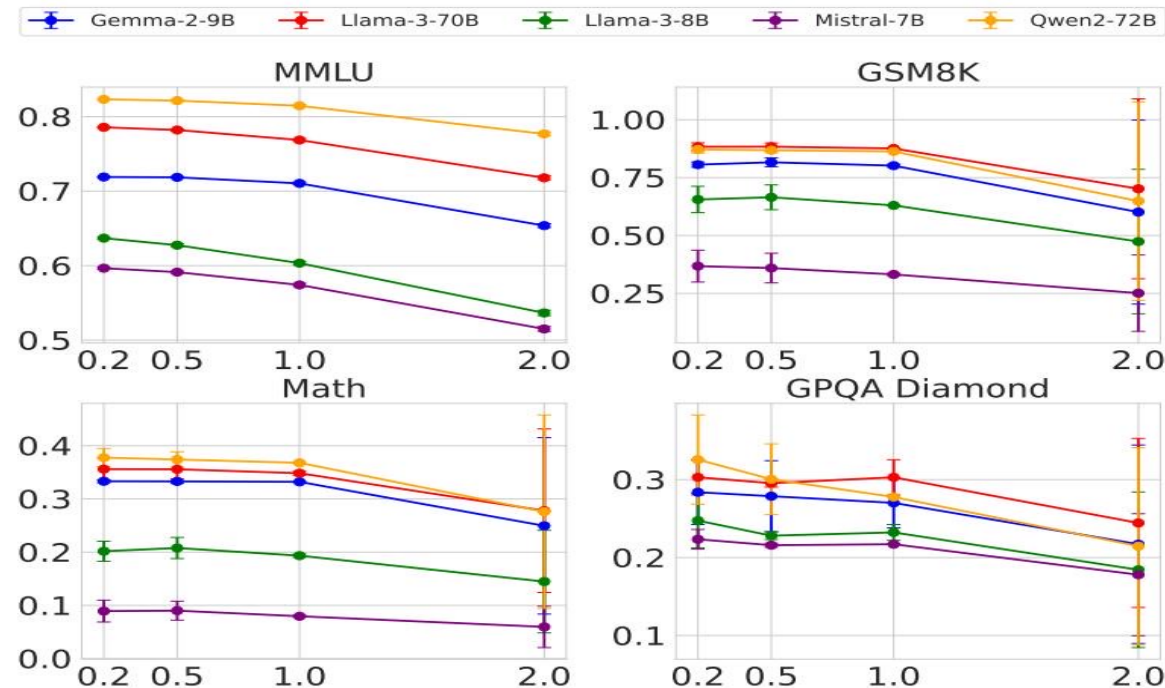
Original vs. FP8-quantized scores across four benchmarks for five model families. Differences often fall within error bars.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Key insight: For most models, FP8 scores fall within the error bars of the original — especially on MMLU and GSM8K — making quantization an attractive, nearly undetectable substitution.

Higher Temperature Amplifies

Benchmark accuracy vs. temperature (0.2, 0.5, 1.0, 2.0) for five models across four tasks. At T = 2.0, degradation is universal and the substitution signal is obscured by noise.



What the data shows

- At low temperature (0.2–0.5), original and FP8 models track closely.
- At T = 2.0, all models degrade — the gap narrows or becomes noisy.
- Temperature scaling is not a reliable discriminator; it adds variance without sharpening the substitution signal.

Log Probability Fingerprinting Fails

The same Llama-3-8B checkpoint produces different token-level log probabilities across software frameworks and GPU hardware

Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why fingerprinting breaks

- Software drift: transformers 4.40 vs. 4.50, vLLM 0.6.2 vs. 0.8.2 alter log-prob outputs.
- Hardware drift: A100 vs. H100 changes numerical paths.
- Auditor cannot distinguish legitimate infrastructure variance from actual model substitution.
- Result: log-probability fingerprinting is unreliable in production.

Randomized Routing at Low Rates

Provider routes only a fraction p of queries to the cheaper model. As $p \rightarrow 0$, the mixed distribution converges to the specified model, requiring exponentially more samples.

Mixed Distribution

$$P_{\text{mixed}}(y \mid x) = (1 - p) \cdot P_{\text{spec}}(y \mid x) + p \cdot P_{\text{sub}}(y \mid x)$$

As p decreases:

- $P_{\text{mixed}} \rightarrow P_{\text{spec}}$
- KL divergence shrinks
- Required sample size grows super-linearly
- Detection becomes economically infeasible

Randomized Substitution

Route only fraction p of traffic to the cheaper model. The auditor sees a distribution arbitrarily close to the genuine one.

Benchmark Evasion

Detect known benchmark prompts and route them to the genuine model while serving substitutes for all regular traffic.

Consequence: software-only statistical detection requires query budgets that are economically prohibitive.

Software-Only Auditing Methods Are Fundamentally Unreliable

A comparison matrix of the three primary failure modes undermining software-based detection.

Method Type	Core Weakness	Practical Limitation
Text-based statistical tests (e.g., MMD)	Require impractically large query budgets; fail against subtle or randomized substitutions	Query volume makes real-time or continuous auditing economically infeasible
Log-probability fingerprinting	Defeated by production-level nondeterminism across software versions and GPU hardware	Cannot distinguish legitimate infrastructure variance from actual model substitution
Adversarial countermeasures	Randomized routing and benchmark evasion actively undermine all detection signals	Provider controls the serving layer; auditor has no ground-truth reference in the black-box setting

Unified conclusion

No software-only method can simultaneously handle quantization-level similarity, production nondeterminism, and active adversarial evasion. The detection problem is fundamentally harder than the substitution problem.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware-secured enclaves that cryptographically attest to the exact model weights and inference code being executed.

Provable

Cryptographic attestation proves specific weights are loaded and correct code is running.

Efficient

Only modest performance overhead compared to ZKPs, which are computationally prohibitive for large models.

Robust

Immune to software nondeterminism and adversarial evasion because verification is hardware-rooted.

Practical Deployment Path

NVIDIA Confidential Computing on H100 GPUs

- GPU boots into a measured, attested state
- Model weights are loaded inside the TEE
- Inference runs with integrity guarantees
- Client receives attestation report before sending data
- Unlike ZKPs: practical overhead, deployable today

TEEs shift the trust model from 'trust the provider' to 'trust the hardware attestation' — a fundamentally stronger guarantee.

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 1 Model substitution is economically motivated: providers can cut GPU costs by serving quantized, smaller, or different models while charging for premium access.
- 2 FP8 quantization barely affects benchmarks: for most model families, quantized scores fall within the error bars of the original, making detection by benchmark comparison nearly impossible.
- 3 Inference nondeterminism kills log-prob fingerprinting: software version and GPU hardware differences produce variance that dwarfs the substitution signal.
- 4 Randomized substitution defeats statistical tests: at low mixing rates, the required query budget grows super-linearly, rendering continuous auditing infeasible.
- 5 TEEs are the only currently viable path: hardware attestation provides provable, efficient, and robust model integrity without trusting the provider's software stack.

Call to action: Adopt hardware attestation standards for LLM APIs and demand TEE-backed integrity reports from your providers.